

Honors Complex Analysis

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Contents

1	Introduction to complex numbers	2
1.1	Basic definitions	2
1.2	Fundamental theorem of algebra	3
2	Holomorphic functions	4
2.1	Cauchy-Riemann equations	4
2.2	Examples of holomorphic functions	5

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1 Introduction to complex numbers

1.1 Basic definitions

Definition 1. The set of complex numbers, $\mathbb{C}$, is defined by

$$\mathbb{C} := \{x + yi | x, y \in \mathbb{R}\},$$

where $i^2 = -1$.

Like real numbers, $\mathbb{C}$ is equipped with basic operations. Let $z = a + bi, w = c + di$, then

- $z + w = (a + bi) + (c + di) = (a + c) + (b + d)i$,
- $zw = (a + bi)(c + di) = (ac - bd) + (bc + ad)i$.
- We have $|z \cdot w| = |z| \cdot |w|$. Verify it yourself.
- Suppose $(a, b) \neq (0, 0)$, then the *inverse* of z is defined by

$$z^{-1} = \frac{1}{z} = \frac{1}{a + bi} = \frac{(a - bi)}{(a + bi)(a - bi)} = \frac{1}{a^2 + b^2}(a - bi).$$

- The *complex conjugate* of $z = a + bi$ is denoted by $\bar{z} = a - bi$, the *norm* of $z = a + bi$ is defined by $|z| = \sqrt{a^2 + b^2}$, hence

$$z \cdot \bar{z} = |z|^2.$$

We may also represent a complex number using polar coordinates:

$$z = a + bi = r(\cos \theta + i \sin \theta),$$

where $r = \sqrt{a^2 + b^2}$, $\theta = \arctan(b/a)$. We define $\arg(z) = \theta$.

Polar representation is useful when multiplying complex numbers. Suppose $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$\begin{aligned} z_1 \cdot z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 \left((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \right) \\ &= r_1 r_2 \left(\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2) \right). \end{aligned}$$

That is, we have

$$\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2).$$

Theorem 1. $\cos \theta + i \sin \theta = e^{i\theta}$.

When $\theta = \pi$, then we have the famous Euler's formula:

$$e^{i\pi} + 1 = 0.$$

The number $e \approx 2.7182818 \dots$ is an irrational number defined through the limit

$$e = \lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} \right)^N.$$

One can also show that

$$\begin{aligned}
e^x &= \left(\lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} \right) \right)^x \\
&= \lim_{N \rightarrow \infty} \left(1 + \frac{1}{N} \right)^{Nx} \text{ by Monotone convergence theorem,} \\
&= \lim_{n \rightarrow \infty} \left(1 + \frac{x}{N} \right)^N \text{ by Bernoulli inequality.}
\end{aligned}$$

Hence we have

$$e^{i\theta} = \lim_{N \rightarrow \infty} \left(1 + \frac{i\theta}{N} \right)^N.$$

As $N \rightarrow \infty$, small angle approximation yields

$$\begin{aligned}
1 + i\frac{\theta}{N} &\approx \cos\left(\frac{\theta}{N}\right) + i\sin\left(\frac{\theta}{N}\right), \\
\left(1 + i\frac{\theta}{N}\right)^N &\approx \left(\cos\left(\frac{\theta}{N}\right) + i\sin\left(\frac{\theta}{N}\right)\right)^N.
\end{aligned}$$

1.2 Fundamental theorem of algebra

Theorem 2. *If $f(z) = a_n z^n + \dots + a_1 z + a_0$ is a polynomial with complex coefficients, then $\exists z \in \mathbb{C}$ such that $f(z) = 0$.*

First proof. If $|z| \gg 0$, $|z| = R > 1$, then $|a_n z^n| = |a_n| \cdot R^n$,

$$\begin{aligned}
|a_{n-1} z^{n-1} + \dots + a_1 z + a_0| &\leq |a_{n-1}| \cdot R^{n-1} + \dots + |a_1| \cdot R + |a_0| \\
&\leq R^{n-1} \cdot (|a_0| + |a_1| + \dots + |a_{n-1}|).
\end{aligned}$$

Let z_0 be a point inside the open disc $|z| = R$ such that $|f(z_0)|$ is minimized. By transforming z_0 to 0, we may assume $|f(0)|$ is the minimum, and $|f(0)| = |a_0|$. Now suppose after transformation, the polynomial is

$$f(z) = a_0 + a_k z^k + \dots + a_n z^n = a_0 + a_k z^k (1 + a_{k+1} z + \dots + a_n z^{n-k})$$

where a_k is the first non-zero coefficients. Choose $z = \left(-\frac{a_0}{a_k}\right)^{1/k} \epsilon$ for some small $\epsilon > 0$, then

$$\begin{aligned}
f(z) &= a_0 - a_0 \epsilon^k \left(1 + \frac{a_{k+1}}{a_k} z + \dots + \frac{a_n}{a_k} z^{n-k} \right) \\
&\approx a_0 (1 - \epsilon^k),
\end{aligned}$$

which implies $|f(z)| < |f(0)|$, a contradiction to the minimality of $|f(0)|$, unless $a_0 = 0$. Hence in original polynomial $f(z_0) = 0$, which is a root. $\square$

Topological proof. Wlog, suppose $f(z) = z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$. When $|z|$ is large, $f(z)$ is dominated by z^n , and

$$|f(z) - z^n| = |a_{n-1} z^{n-1} + \dots + a_1 z + a_0| < C \cdot |z|^{n-1},$$

where C is some constant not depending on z . Let $|z| = R$ be the circle with radius R , then z^n is the circle with radius R^n and winding n times. The remaining term $a_{n-1}z^{n-1} + \dots + a_0$ will perturbate around $z^n = R^n$. When $R = 0$, the path is concentrated around a_0 . For some R , the path must cross 0, indicating a root. $\square$

Using Liouville's theorem. We claim that any bounded differentiable function $f : \mathbb{C} \rightarrow \mathbb{C}$ is constant (Liouville's theorem). Suppose $p(z) \in \mathbb{C}[z]$ is a polynomial with no roots, then let $f(z) = \frac{1}{p(z)}$, then

$$\frac{d}{dz} \frac{1}{p(z)} = -\frac{p'(z)}{p^2(z)},$$

and by Liouville's theorem, $f(z)$ is a constant, so is $p(z)$. $\square$

2 Holomorphic functions

2.1 Cauchy-Riemann equations

Definition 2. A function $f(z) : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic if it has a well defined derivative, i.e.,

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} \text{ exists and is well defined.}$$

Let $f(z) = f(x+iy) = u(x, y) + iv(x, y)$, and suppose f is holomorphic. We compute the derivative $f'(z)$ in two ways: First, let $h \in \mathbb{R}$:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} &= \lim_{h \rightarrow 0} \frac{u(x+h, y) + iv(x+h, y) - u(x, y) - iv(x, y)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x+h, y) - u(x, y)}{h} + i \lim_{h \rightarrow 0} \frac{v(x+h, y) - v(x, y)}{h} \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}. \end{aligned} \tag{2.1}$$

Second, let $h \in \mathbb{R}$:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z+ih) - f(z)}{ih} &= \lim_{h \rightarrow 0} \frac{u(x, y+ih) + iv(x, y+ih) - u(x, y) - iv(x, y)}{ih} \\ &= \lim_{h \rightarrow 0} \frac{u(x, y+ih) - u(x, y)}{ih} + i \lim_{h \rightarrow 0} \frac{v(x, y+ih) - v(x, y)}{ih} \\ &= \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}. \end{aligned} \tag{2.2}$$

Since f is holomorphic, by (2.1) and (2.2), we have the *Cauchy-Riemann equation*:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}}. \tag{2.3}$$

If $f = (u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, the Jacobian of f is defined as

$$\mathcal{J}_f = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \stackrel{(2.3)}{=} \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \quad a, b \in \mathbb{R}.$$

Note that if we let $z = a + bi$ and define a map T by $w \mapsto (a + bi)w$ where $w \in \mathbb{C}$. Take the basis of $\mathbb{C}$ as $(1, 0), (0, i)$, then the transformation matrix is given by

$$T(w) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} w.$$

Think of $z = \cos \theta + i \sin \theta$, we hence have the transformation matrix:

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

which is the counter-clockwise rotation by θ .

2.2 Examples of holomorphic functions

We investigate some holomorphic functions:

① $f(z) = z^n$. We have that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} &= \lim_{h \rightarrow 0} \frac{(z+h)^n - z^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sum_{k=0}^n \binom{n}{k} z^{n-k} h^k}{h} \\ &= \lim_{h \rightarrow 0} n z^{n-1} + \binom{n}{2} h z^{n-2} + \dots + h^{n-1} \\ &= n z^{n-1}. \end{aligned}$$

We have $f'(z) = n z^{n-1}$.

② If $f(z), g(z)$ are holomorphic, then so is

$$\lambda_1 f(z) + \lambda_2 g(z), \quad \lambda_1, \lambda_2 \in \mathbb{C},$$

and

$$(\lambda_1 f(z) + \lambda_2 g(z))' = \lambda_1 f'(z) + \lambda_2 g'(z).$$

Based on this property, $f(z) \in \mathbb{C}[z]$ is holomorphic.

③ Power series: A power series is an infinite sum

$$\sum_{n=0}^{\infty} a_n z^n, \quad a_n \in \mathbb{C},$$

we say the series converges at $z_0 \in \mathbb{C}$ if

$$\lim_{N \rightarrow \infty} \sum_{n=0}^N a_n z_0^n \text{ exists.}$$

We will see later that, if the power series converges absolutely in $\Omega \subset \mathbb{C}$, then it defines a holomorphic function.